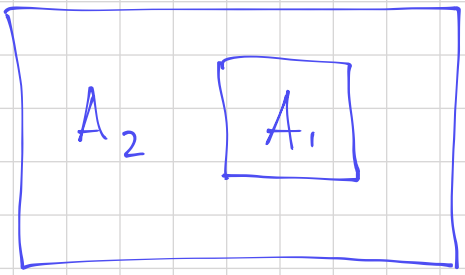
Canonical ensemble



$A_2: E_2, V_2, N_2$

$A_1: E_1, V_1, N_1$

$E = E_1 + E_2 = \text{const}, \quad E_2 \gg E_1$

$$H = H(q_1, q_2, p_1, p_2) = H_1(q_1, p_1) + H_2(q_2, p_2)$$

Combined distribution function of the system $A = A_1 + A_2$:

$$\rho(q, p) = \frac{\delta(E - H_1 - H_2)}{\int d\Gamma \, \delta(E - H_1 - H_2)}$$

here $d\Gamma = d^{3N}\vec{x} \, d^{3N}\vec{p}$, $\Omega(E) = \int d\Gamma \, \delta(E - H)$

density of states

$\Rightarrow$ distribution function of small system

$$\rho_1(q_1, p_1) = \int d\Gamma_2 \, \rho(q, p) = \frac{\int d\Gamma_2 \, \delta(E - H_1 - H_2)}{\Omega(E)}$$

$$= \frac{\Omega_2(E - H_1)}{\Omega(E)}$$

Now expand in small $E_1 \ll E$:

$$\ln \Omega_2(E - H_1) = \ln \Omega_2(E) - \left(\frac{\partial \ln \Omega_2}{\partial E_2}\right)_{E_2 = E} \cdot H_1 + \ldots$$

but $$\frac{\partial \ln \Omega_2(E)}{\partial E} = \frac{1}{k_B} \frac{\partial S_2}{\partial E} = \frac{1}{k_B T} \equiv \beta$$

$$\Rightarrow \Omega_2(E - H_1) = \Omega_2(E) \, e^{-H_1/k_B T}$$

$$\Rightarrow \rho_1 = \frac{\Omega_2(E) \, e^{-\beta H_1}}{\Omega(E)},$$ since $\frac{\Omega_2(E)}{\Omega(E)}$ is H_1-independent,

$$\rho_1 = e^{-\beta H_1} \Big/ \int d\Gamma_1 \, e^{-\beta H_1}$$

Canonical partition function:

$$Z_N = \int \frac{d^{3N}q \, d^{3N}p}{h^{3N} N!} e^{-\beta H(q,p)}$$

Let us consider $\frac{\partial}{\partial \beta} \ln Z_N = \frac{1}{Z_N} \frac{\partial Z_N}{\partial \beta} = \frac{1}{Z_N} \int d\Gamma (-H) e^{-\beta H}$

$$= -\langle H \rangle$$

but $\langle H \rangle = E = -\frac{\partial}{\partial \beta} \ln Z_N$

Next, $F = E - TS$

$$E = F + TS = F - T\left(\frac{\partial F}{\partial T}\right)_{V,N} = -T^2 \frac{\partial}{\partial T}\left(\frac{F}{T}\right)$$

$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N} \quad \text{thermodynamic relation}$$

$$\Rightarrow E = \frac{\partial}{\partial \beta}(\beta F) \quad \text{since} \quad \frac{\partial}{\partial T} = \frac{\partial}{\partial \beta}\frac{\partial \beta}{\partial T} = -\frac{1}{T^2}\frac{\partial}{\partial \beta}$$

Therefore we obtained

$$-\frac{\partial}{\partial \beta} \ln Z_N = \frac{\partial}{\partial \beta}(\beta F) \Rightarrow \boxed{Z_N = e^{-\beta F}}$$

$\Rightarrow$ can get thermodynamic properties from the partition function!

$$F = -k_B T \ln Z_N \Rightarrow S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}, \; P = -\left(\frac{\partial F}{\partial V}\right)_{T,N}$$

$$\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V}$$

Heat capacity $C_V = \left(\frac{\partial E}{\partial T}\right)_{V,N} = \frac{\partial}{\partial T}\left[-T^2 \frac{\partial}{\partial T}\left(\frac{F}{T}\right)\right]$

$$= -T \frac{\partial^2 F}{\partial T^2} = T \frac{\partial S}{\partial T}$$

Ideal gas $H = \sum_1^N \frac{\vec{p}_i^{\,2}}{2m}$

$$Z_N(T,V) = \int \frac{d^{3N}q \, d^{3N}p}{h^{3N} N!} e^{-\beta \sum \frac{\vec{p}_i^{\,2}}{2m}} =$$

$$= \frac{V^N}{N!} \left(\int_{-\infty}^{\infty} \frac{dp}{h} e^{-\beta \frac{p^2}{2m}} \right)^{3N}$$

$$p = \sqrt{\frac{2m}{\beta}} x \Rightarrow \sqrt{\frac{2m}{\beta}} \frac{1}{h} \underbrace{\int_{-\infty}^{\infty} dx \, e^{-x^2}}_{\sqrt{\pi}} = \frac{1}{h}\sqrt{2\pi m k_B T}$$

$$Z_N = \frac{V^N}{N!} \left(\frac{\sqrt{2\pi m k_B T}}{h} \right)^{3N}$$

$$F = -k_B T \left\{ \ln \frac{V^N}{N!} + N \ln \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} \right\}$$

$$\ln N! = N \ln N - N$$

$$F = -k_B T N \left\{ \ln \frac{V}{N} \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} + 1 \right\}$$

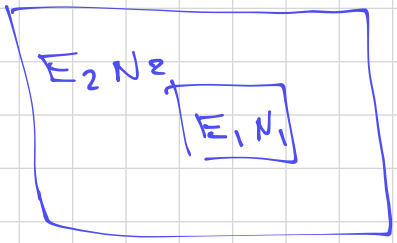
$$S = -\frac{\partial F}{\partial T} = N k_B \left\{ \ln \frac{V}{N} \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} + \frac{5}{2} \right\}$$

$$\Rightarrow C_V = T \left(\frac{\partial S}{\partial T} \right)_{V,N} = N k_B T \underbrace{\frac{\partial}{\partial T} \left\{ \ln T^{3/2} + \dots \right\}}_{\frac{3}{2T}} = \frac{3}{2} N k_B !$$

Much easier than microcanonical calculation even though the final answer is the same.

But $\rightarrow$ in canonical ensemble Z_N <u>factorizes</u> (which makes calculation very simple)

Grand canonical ensemble: allow <u>exchange</u> of particles



$$\rho_1(q_1, p_1) = \frac{\int d\Gamma_2 \, \delta(E - H_1 - H_2)}{\int d\Gamma \, \delta(E - H_1 - H_2)} = \frac{\Omega_2(E - H_1, N - N_1)}{\Omega(E, N)}$$

$$S_2(E - E_1, N - N_1) = k_B \ln \Omega_2(E - E_1, N - N_1)$$

$$\approx \underbrace{k_B \ln \Omega_2(E, N)}_{S_2(E,N)} - E_1 \left(\frac{\partial S_2}{\partial E_2}\right)_{N_2} \Bigg|_{\substack{E_2 = E \\ N_2 = N}}$$

$$- N_1 \left(\frac{\partial S_2}{\partial N_2}\right)_{E_2} \Bigg|_{\substack{E_2 = E \\ N_2 = N}}$$

but $\left(\frac{\partial S_2}{\partial E_2}\right)_{N_2} = \frac{1}{T}$

$$\left(\frac{\partial S_2}{\partial N_2}\right)_{E_2} = -\frac{\mu}{T}$$

$$\Rightarrow S_2(E - E_1, N - N_1) = S_2(E, N) - \frac{E_1}{T} + \frac{\mu N_1}{T}$$

$$\Rightarrow \rho_1 \sim e^{-\beta(H_1 - \mu N_1)}$$

$\Rightarrow$ grand partition function

$$Z(T, V) = \sum_{N_1=0}^{\infty} \int \frac{d^{3N_1} p \, d^{3N_1} q}{h^{3N_1} \, N_1!} e^{-\frac{H_1 - \mu N_1}{k_B T}}$$

$$= \sum_{N_1=0}^{\infty} \left(e^{\beta\mu}\right)^{N_1} Z_{N_1}(T, V) \; ; \; z = e^{\beta\mu}$$

fugacity

Quantum systems

Partition function $Z = \sum_\alpha e^{-\beta E_\alpha} = e^{-\beta F}$

sum over all states of the system (this includes degeneracies, i.e. different states with the same energy)

Average energy $E = \frac{1}{Z} \sum_\alpha E_\alpha e^{-\beta E_\alpha} = -\frac{\partial}{\partial \beta} \ln Z$

Chemical potential is defined by $\mu = \frac{\partial F}{\partial N}$.

Grand patition function $Z = \sum_{N,\alpha} e^{-\beta(E_\alpha - \mu N)}$

Average number of particles $N = -k_B T \frac{\partial}{\partial \mu} \ln Z$.